Example 2.22 (i) Add $3\sqrt{7}$ and $5\sqrt{7}$. Check whether the sum is rational or irrational. (ii) Subtract $4\sqrt{5}$ from $7\sqrt{5}$. Is the answer rational or irrational?

Solution

(i) $3\sqrt{7}+5\sqrt{7} = (3+5)\sqrt{7} = 8\sqrt{7}$. The answer is irrational.

(ii) $7\sqrt{5}-4\sqrt{5} = (7-4)\sqrt{5} = 3\sqrt{5}$. The answer is irrational.

Example 2.23 Simplify the following:

(i) $\sqrt{63}-\sqrt{175}+\sqrt{28}$ (ii) $2\sqrt[3]{40}+3\sqrt[3]{625}-4\sqrt[3]{320}$

Solution

(i) $\sqrt{63}-\sqrt{175}+\sqrt{28} = \sqrt{9\times7}-\sqrt{25\times7}+\sqrt{4\times7}$

$= 3\sqrt{7}-5\sqrt{7}+2\sqrt{7}$

$= \left(3\sqrt{7}+2\sqrt{7}\right)-5\sqrt{7}$

$= 5\sqrt{7}-5\sqrt{7}$

$= 0$

(ii) $2\sqrt[3]{40}+3\sqrt[3]{625}-4\sqrt[3]{320}$

$= 2\sqrt[3]{8\times5}+3\sqrt[3]{125\times5}-4\sqrt[3]{64\times5}$

$= 2\sqrt[3]{2^3\times5}+3\sqrt[3]{5^3\times5}-4\sqrt[3]{4^3\times5}$

$= 2\times2\sqrt[3]{5}+3\times5\sqrt[3]{5}-4\times4\sqrt[3]{5}$

$= 4\sqrt[3]{5}+15\sqrt[3]{5}-16\sqrt[3]{5}$

$= (4+15-16)\sqrt[3]{5} = 3\sqrt[3]{5}$

(ii) Multiplication and division of surds

Like surds can be multiplied or divided by using the following rules:

Multiplication property of surds	Division property of surds
(i) $\sqrt[n]{a}\times\sqrt[n]{b}=\sqrt[n]{ab}$	(iii) $\dfrac{\sqrt[n]{a}}{\sqrt[n]{b}}=\sqrt[n]{\dfrac{a}{b}}$
(ii) $a\sqrt[n]{b}\times c\sqrt[n]{d}=ac\sqrt[n]{bd}$ where $b,d>0$	(iv) $\dfrac{a\sqrt[n]{b}}{c\sqrt[n]{d}}=\dfrac{a}{c}\sqrt[n]{\dfrac{b}{d}}$ where $b,d>0$

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